

MGS 602: IT & Cloud Infrastructure Management

Introduction to AWS IAM

1) Imagine you are an administrator managing the IAM users and groups for a large organization. What are some challenges you might face in assigning and managing permissions for a large number of users and groups?

Response: There can be difficulties like-

As the organization expands, it becomes more challenging to organize and manage permissions for a growing number of users and groups. Monitoring Adjustments can be an issue, more precisely, maintaining record of individuals with permissions and ensuring security and compliance can present a significant challenge. Also, with teams evolving and expanding, it can be challenging to ensure groups are kept current and users are correctly assigned. When it comes to that, finding the appropriate level of access for various teams can be challenging in order to avoid granting excessive or insufficient permissions.

Last but not the least, users may often acquire an excess of permissions beyond their necessity, resulting in potential security hazards.

2) Think about a real-world scenario in which you might need to assign permissions to users and groups in AWS IAM. Describe the process you would follow to ensure that each user has the appropriate level of access to AWS resources while maintaining security.

Response: Here is the procedure I will follow to grant permissions using AWS IAM while still maintaining security.

Initially, I would determine the tasks required for each team to complete. One scenario could involve developers requiring access to EC2 and repositories for code, while analysts require access to databases and data stored in S3. Followed by establishing User Groups i.e. classifying users according to their roles. For example, form categories for "Programmers," "Data Specialists," and "Customer Service Representatives." This makes managing permissions easier by assigning policies to groups rather than to individuals. In addition to this, I will develop policies for each group that provide them with the least amount of permissions required to complete their responsibilities. Developers may have the privilege to control the start/stop function of EC2 instances without having access to billing details.

Next, establishing IAM policies for every group. Developers may have an EC2 policy allowing them to manage instances, whereas analysts might be limited to a read-only policy for S3. Connecting these guidelines to the appropriate groups. In this manner, users are able to take on roles as required, minimizing unneeded access. Multi-factor authentication could be used in order to add additional level of security.

3) How do IAM policies work in AWS IAM, and what are the effects of policies on service access? Can you give an example of a policy and explain its basic structure?

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Response: In AWS IAM, policies are JSON documents that control what actions users, groups, or roles can perform on AWS resources. Each policy specifies the operations allowed or denied (e.g., `ec2:StartInstances`), AWS resources the actions apply to (e.g., EC2 instances in a specific region) and whether the action is Allow or Deny.

Example of a policy to start and stop EC2 instances in the **us-west-2** region:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances"
      ],
      "Resource": "arn:aws:ec2:us-west-2:123456789012:instance/*"
    }
  ]
}
```

Impact:

The effect is that only the specified actions are permitted ("Allow"). The user is allowed to start (`ec2:StartInstances`) and stop (`ec2:StopInstances`) EC2 instances. And the policy applies to all EC2 instances in the us-west-2 region under the account 123456789012.

With this policy, the user can only start and stop EC2 instances in the specified region, but cannot terminate or modify them without additional permissions.

4) In this lab, you were given pre-created IAM users and groups. What are some ways in which you could create your own IAM users and groups, and how might you manage their permissions?

Response: I can create IAM users and groups using:

1. AWS Management Console: by creating users in the "Users" section and groups in the "Groups" section. Attach policies to manage permissions.

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2. AWS CLI: by using commands like ``aws iam create-user`` for users and ``aws iam create-group`` for groups. Manage permissions with ``aws iam attach-user-policy`` or ``aws iam attach-group-policy``.
3. AWS SDKs: by automating user and group creation with programming languages using relevant SDK functions.
4. Infrastructure as Code (IaC): by using AWS CloudFormation or Terraform to define IAM resources in code and manage permissions through IAM policies.

Managing Permissions-

- a) IAM Policies: defining granular permissions for actions on resources.
- b) Managed Policies: using AWS-managed or customer-managed policies.
- c) Group Policies: assigning permissions at the group level for all users in that group.
- d) Permission Boundaries: by controlling maximum permissions for IAM users.

5) What is the difference between a managed policy and an inline policy in AWS IAM, and in what situations might you choose to use one over the other?

Response: In AWS IAM, a managed policy is an independent policy that can be applied to various users, groups, or roles, created by either AWS or the user. It's wonderful to have uniform permissions for numerous identities. On the other hand, an inline policy is directly linked to a specific user, group, or role and is specific to that entity.

I would prefer inline policies for more personalized or short-term access needs whereas I would use managed policies for reusability when I would want to use it in longer-term,

6) What is the difference between a user and a role in AWS IAM?

Response: In AWS IAM, a **User** is an individual with long-term credentials, while a **Role** can be temporarily assumed by trusted entities and uses temporary credentials. Users have enduring access, while roles provide temporary access or permissions across multiple accounts.

7) Can you explain the concept of least privilege and how it relates to AWS IAM?

Response: The idea of least privilege involves giving users and roles only the essential permissions for their job tasks, reducing the chance of unintentional or harmful actions. In AWS IAM, this concept is implemented through the creation of policies that permit only the essential actions on the required resources, preventing users from having excessive access. This increases safety by decreasing the possible areas for attack.

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8) How can you use IAM groups to manage permissions for multiple users, and what are some best practices for using IAM groups?

Response: Using IAM groups in AWS effectively manages permissions for numerous users. Creating groups based on roles or job functions allows you to grant permissions to the group as a whole instead of granting them individually to each user. This simplifies access management by allowing for easy updates or removals from a group when a user changes roles or leaves the organization instead of adjusting permissions for each user separately.

The suggested methods for utilizing IAM groups are-

1. Group by Job Title: Form teams according to similar roles (such as developers, administrators, analysts) for easier management of permissions.
2. Restrict Group Size: Maintain small, specialized groups to prevent wide-ranging access and simplify auditing processes.
4. Frequently Monitor Group Membership: Regularly review group memberships and permissions to make sure they match current roles and duties.
5. Enforce Least Privilege: Make sure that groups are only granted the permissions they need for their designated duties, reducing potential security threats.

By adhering to these methods, you can enhance access control and increase the security of your AWS setup.

9) Provide a brief executive summary of the lab you completed along with screenshots.

Response: In this lab activity, I worked upon AWS Lab and examined Identity and Access Management (IAM) to administer access rights according to job positions.

First of all, three users; **user-1, user-2, and user-3**, were created with no permissions. Post which, three groups were formed: **EC2-Admin, EC2-Support, and S3-Support**. Followed by that, assignments were given to the users in a way-

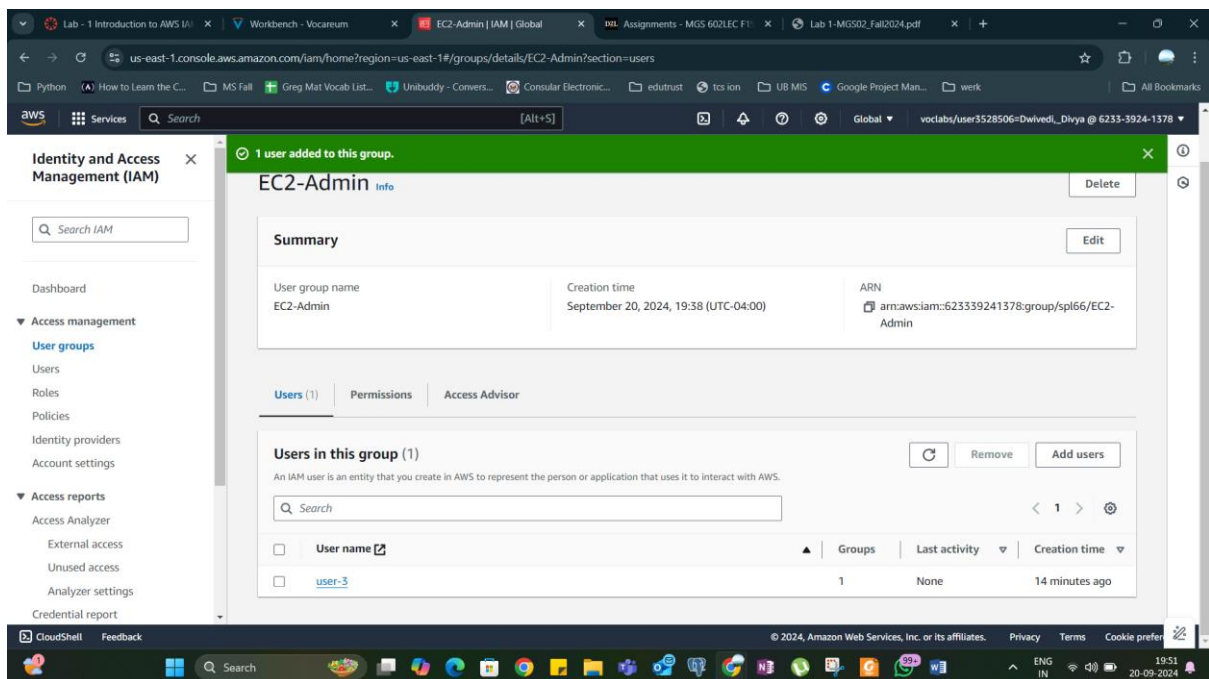
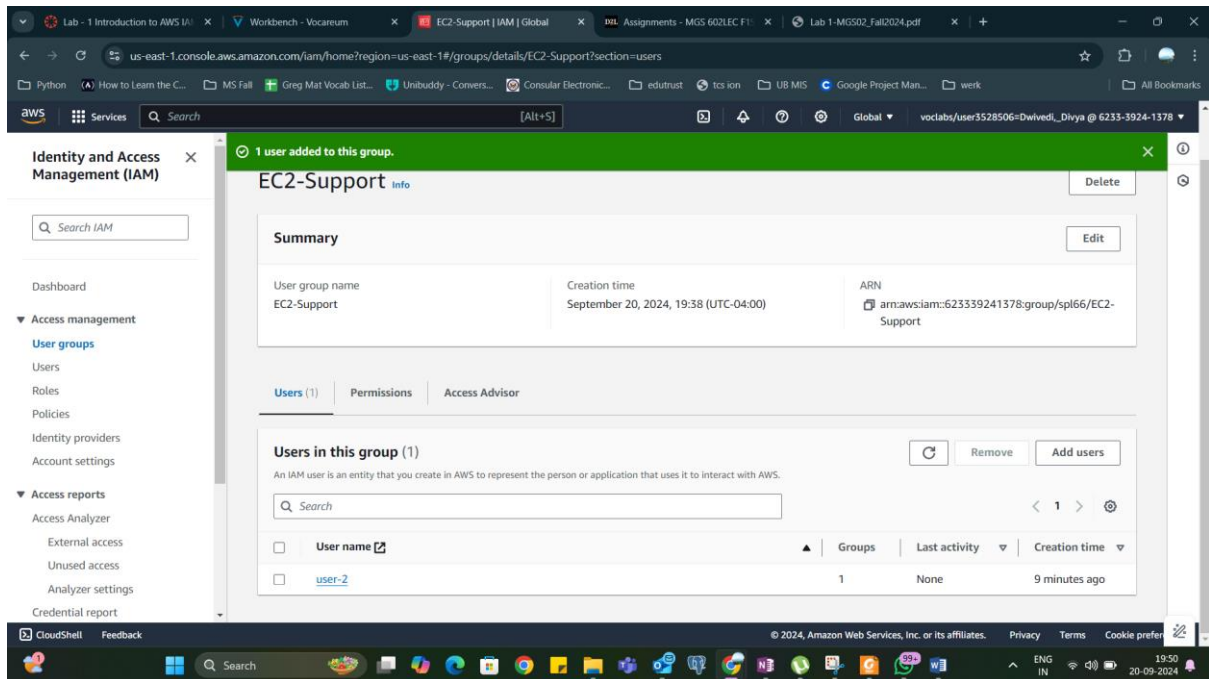
- a. user-1: Included in S3-Support is the ability to view S3 with read-only permission.
- b. user-2: EC2-Support was included for read-only entry to EC2.
- c. user-3: EC2-Admin was given permission to manage EC2 instances.

By effectively adhering the respective services to the respective users, signing in and checking whether the users are added in the respective service, and logging out, I managed to effectively complete the first lab and achieved the score of 100%.

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SCREENSHOTS-



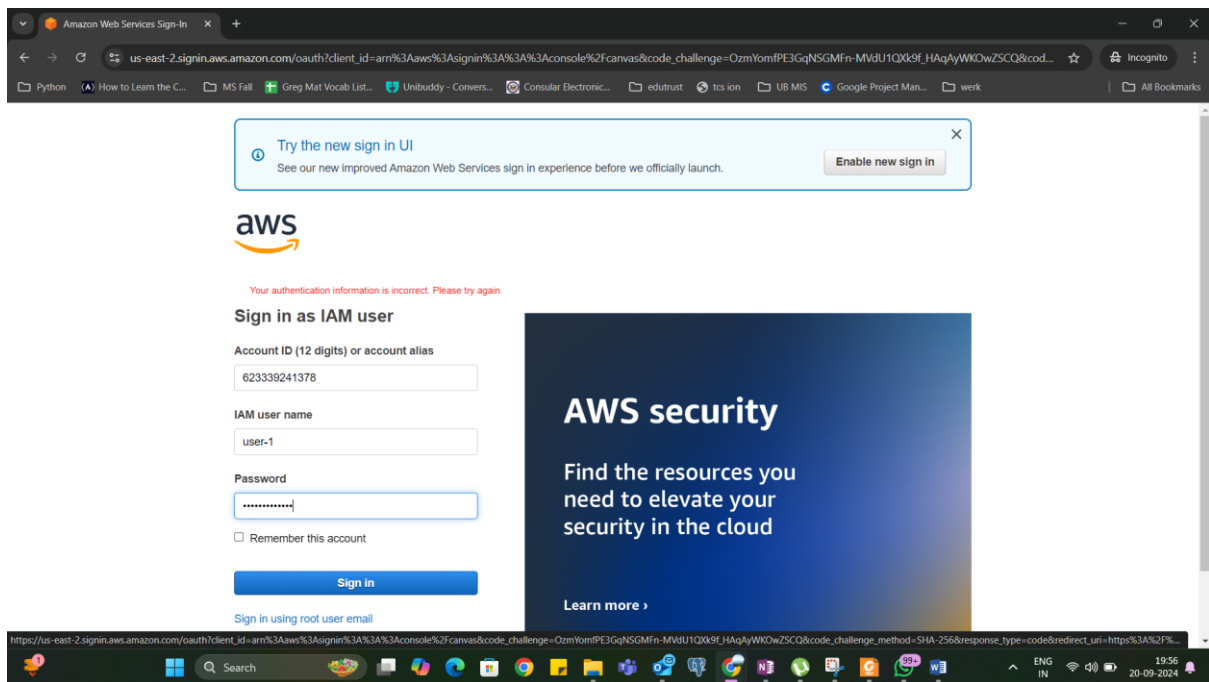
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The screenshot shows the AWS IAM console interface. The left sidebar contains the 'Identity and Access Management (IAM)' menu with options like Dashboard, Access management, Access reports, and Access Advisor. The main content area displays the 'Summary' tab for the 'EC2-Admin' user group. A green notification banner at the top states '1 user added to this group.' The summary section includes fields for 'User group name' (EC2-Admin), 'Creation time' (September 20, 2024, 19:38 (UTC-04:00)), and 'ARN' (arn:aws:iam::623339241378:group/spl66/EC2-Admin). Below the summary, the 'Permissions policies (1)' section is visible, showing a table with one policy: 'EC2-Admin-Policy' of type 'Customer inline' with 0 attached entities. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 19:52 on 20-09-2024.

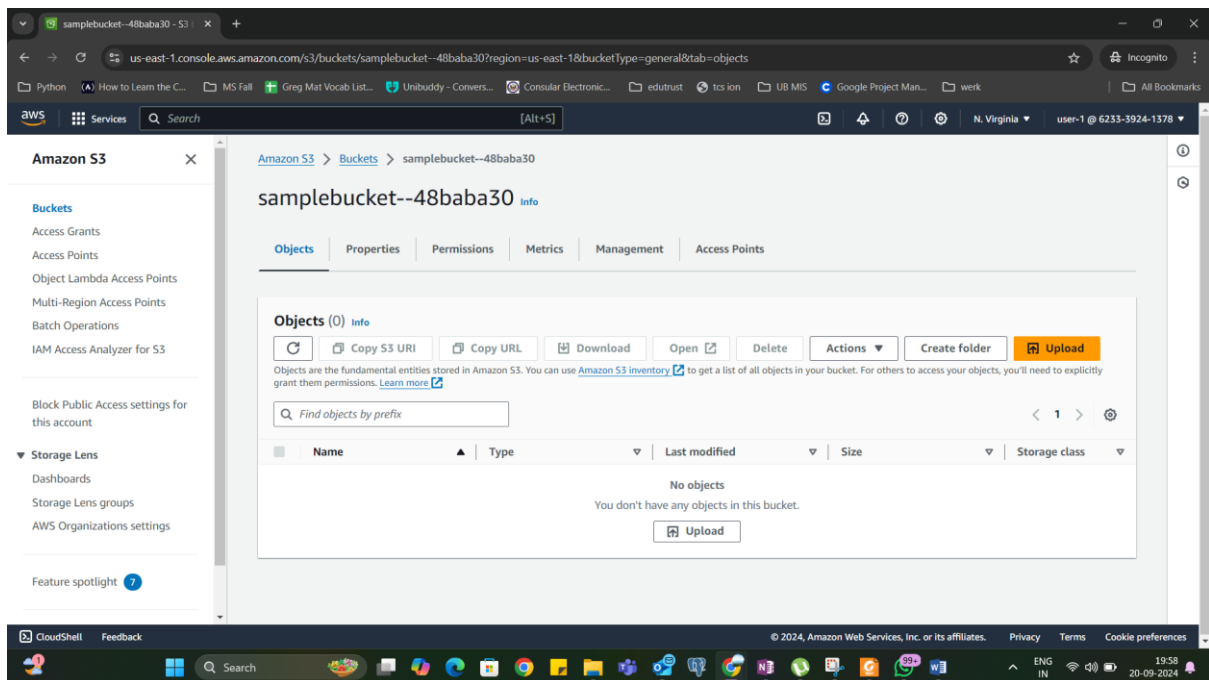
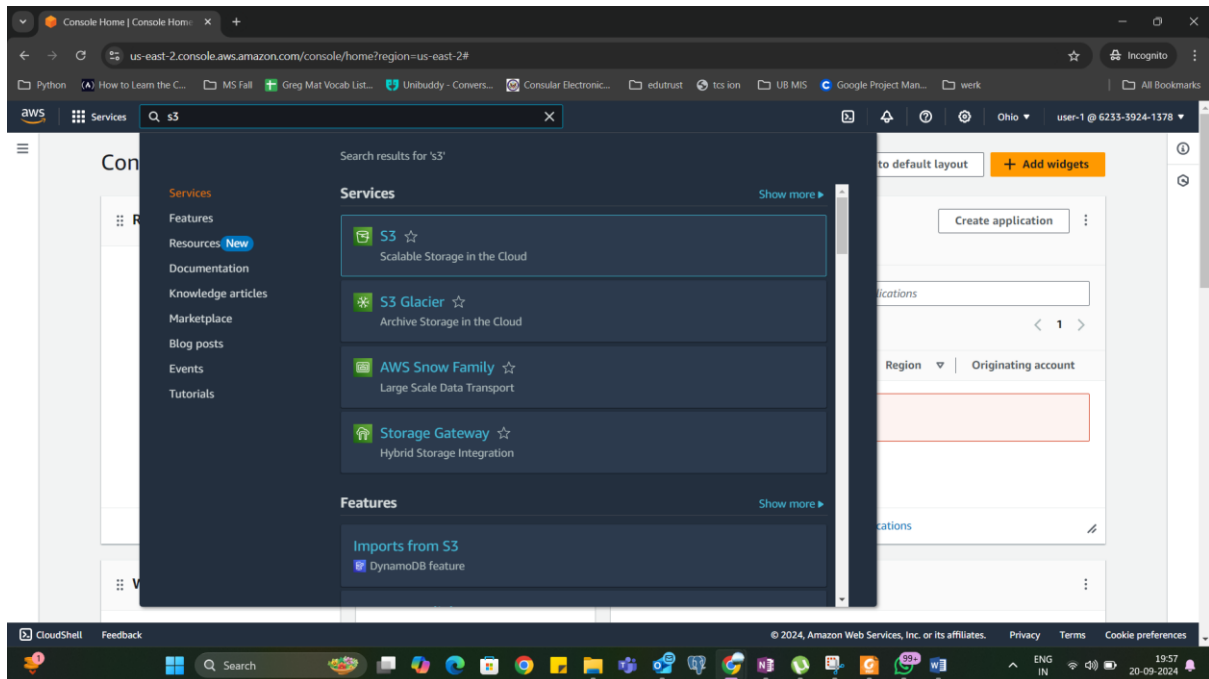
The screenshot shows the AWS IAM console interface for the 'user-1' user. The left sidebar is the same as the previous screenshot. The main content area displays the 'Summary' tab for 'user-1'. A 'Delete' button is visible in the top right corner. The summary section includes fields for 'ARN' (arn:aws:iam::623339241378:user/spl66/user-1), 'Created' (September 20, 2024, 19:37 (UTC-04:00)), 'Console access' (Enabled without MFA), and 'Last console sign-in' (Never). Below the summary, the 'Security credentials' tab is selected, showing 'Access key 1' (AKIAZCIPJ76GROTUDOSPR - Active) and 'Access key 2' (Create access key). The 'Console sign-in' section shows the 'Console sign-in link' (https://623339241378.signin.aws.amazon.com/console) and 'Console password' (Updated 15 minutes ago (2024-09-20 19:38 EDT)). The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 19:53 on 20-09-2024.

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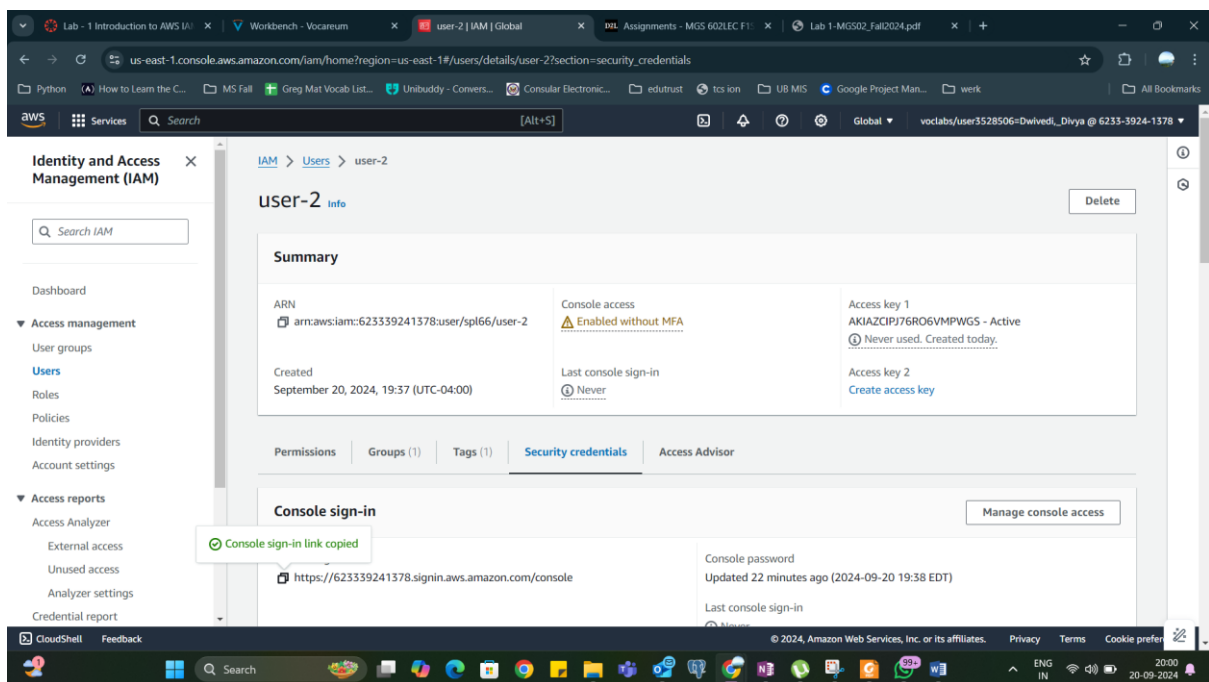
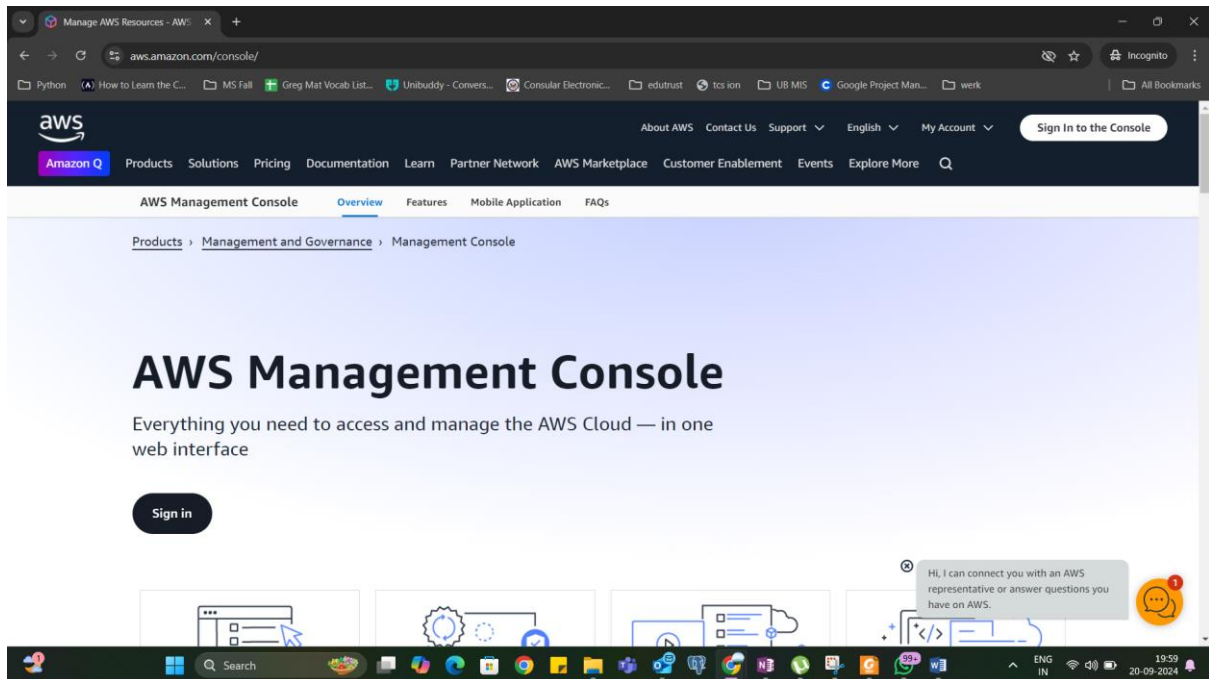
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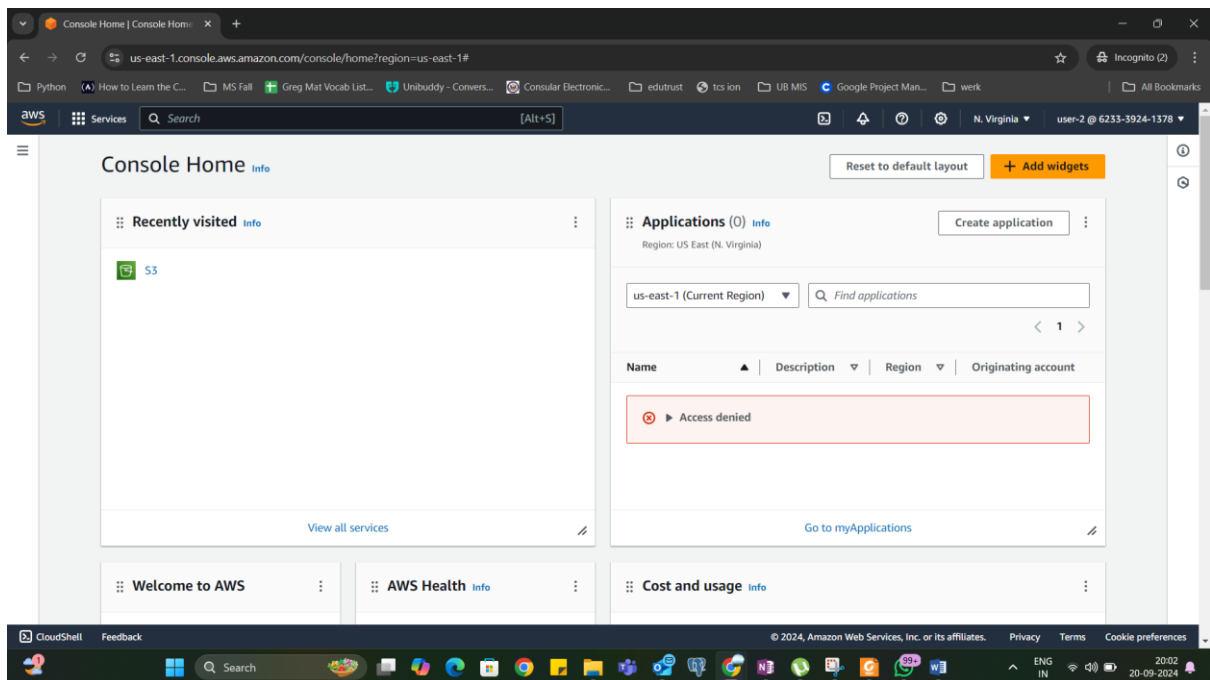
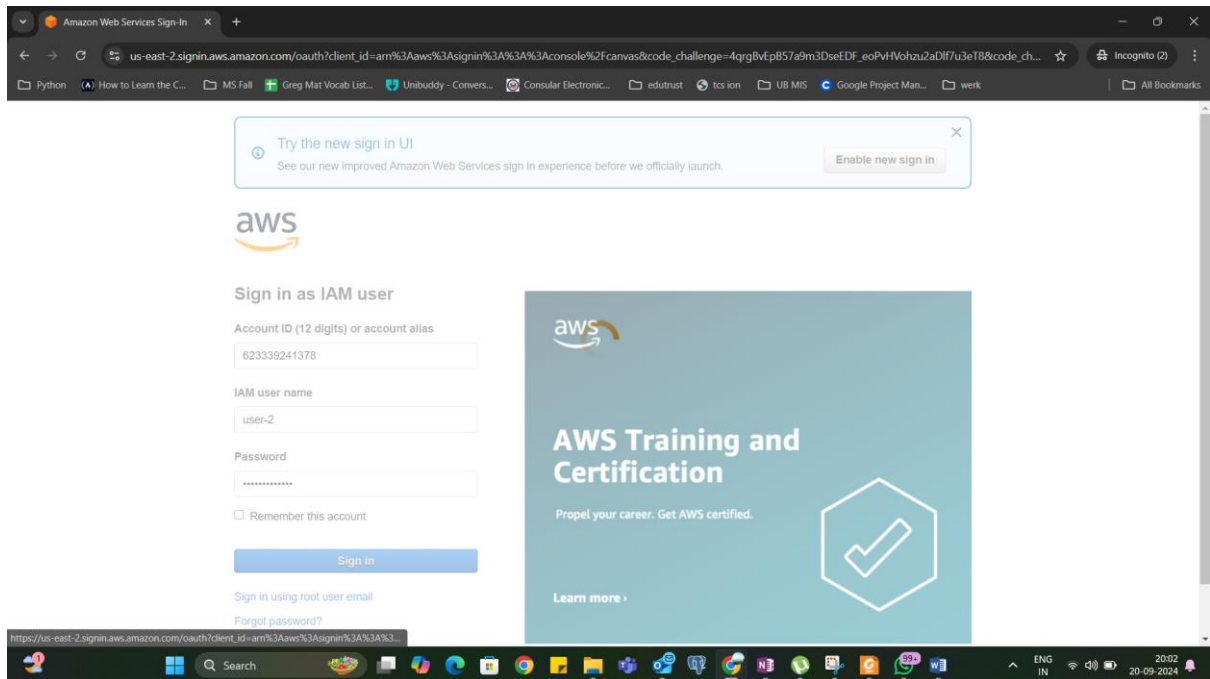
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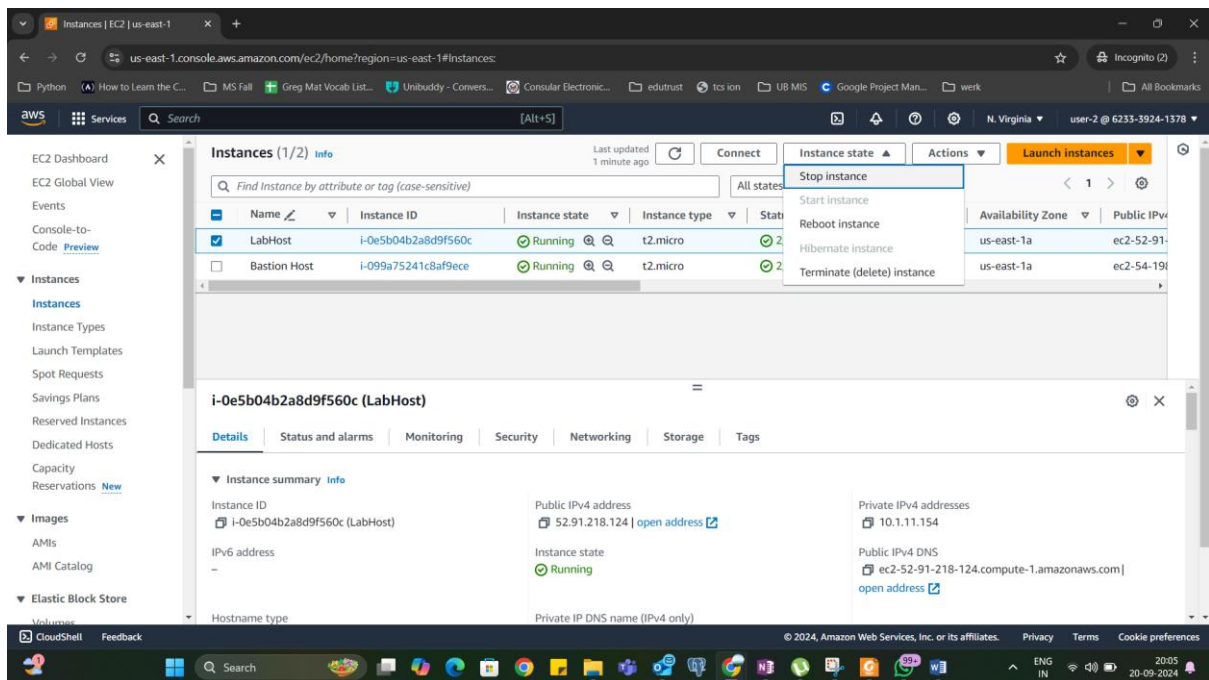
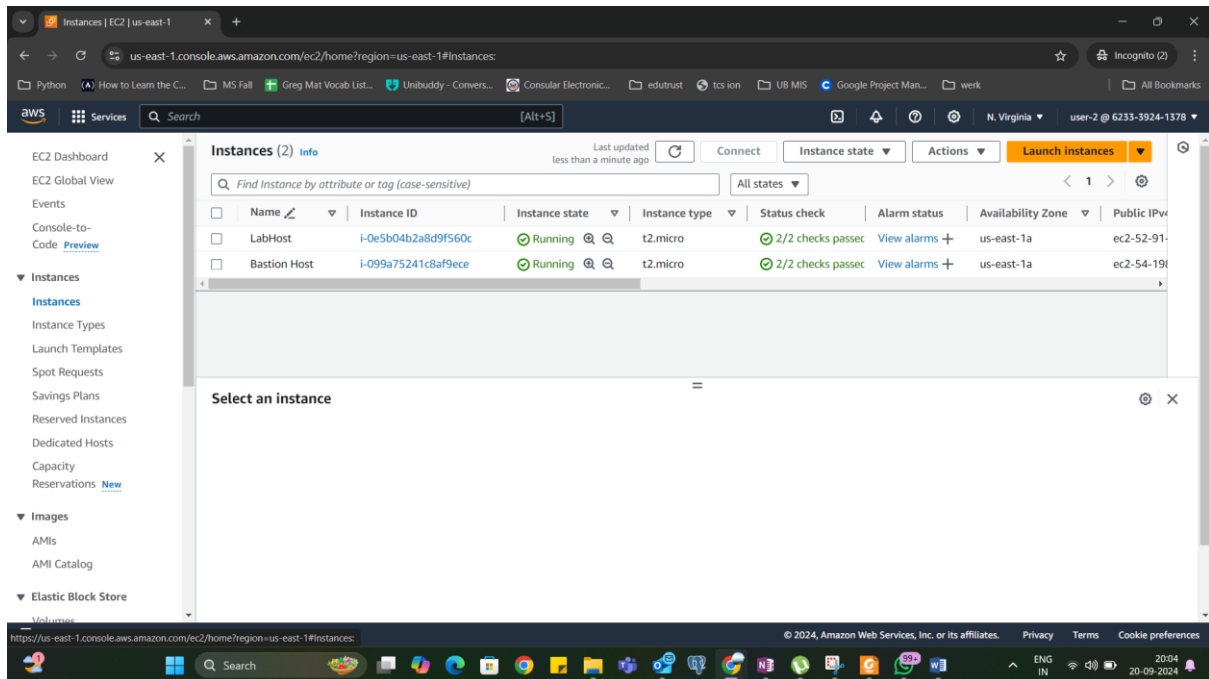
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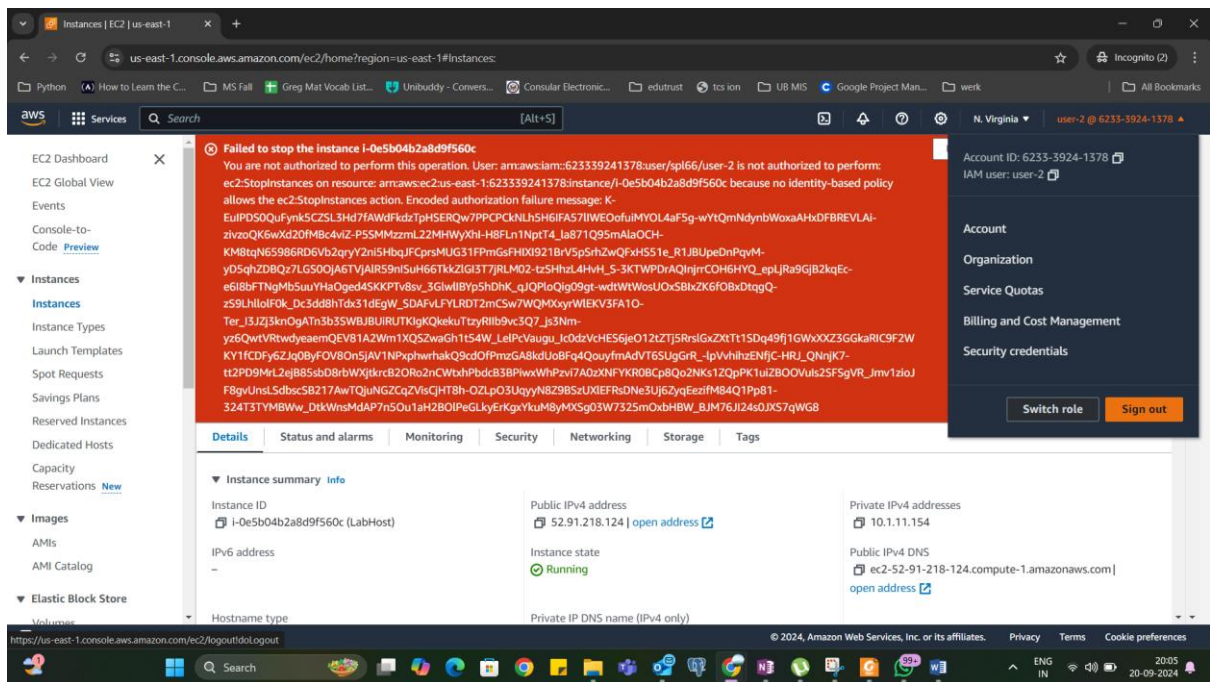
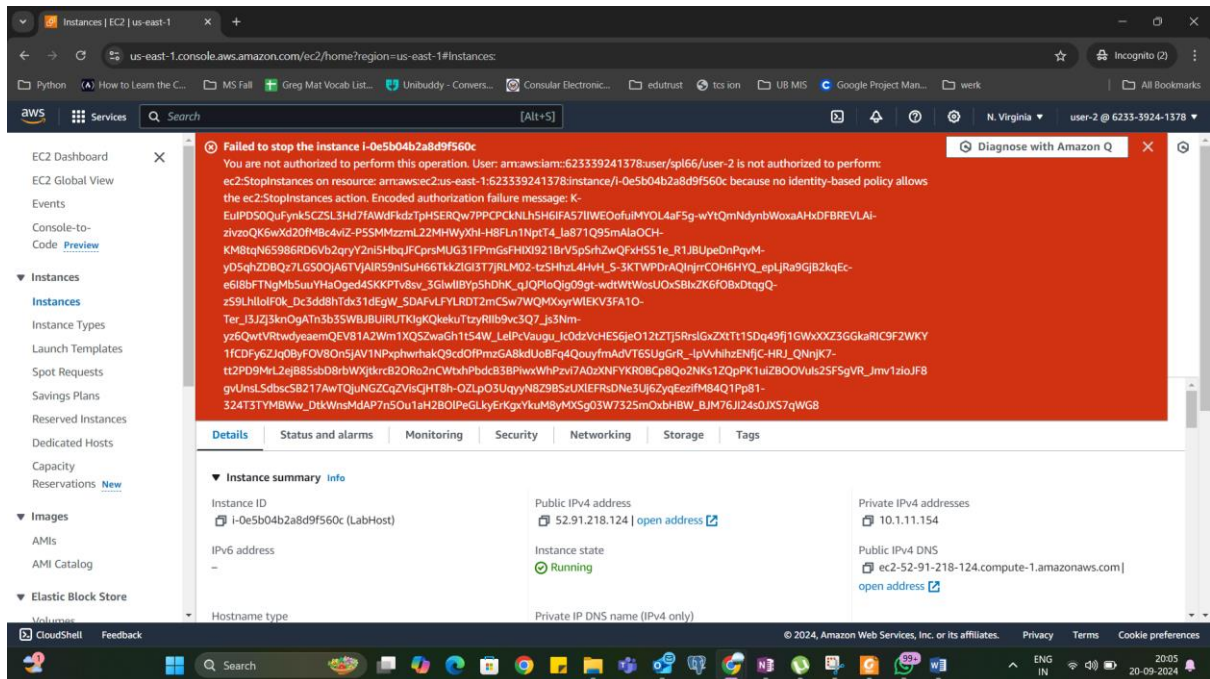
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The screenshot shows the AWS IAM console interface. The left sidebar contains navigation options: Identity and Access Management (IAM), Access management, Access reports, and CloudShell. The main content area displays the details for 'user-3'. The 'Summary' section includes the ARN (arn:aws:iam::623339241378:user/spl66/user-3), Console access status (Enabled without MFA), and Access key 1 (AKIAZCIPJ76GROWVZM5I - Active). The 'Security credentials' tab is selected, showing the 'Console sign-in' section with a link to the console and a 'Manage console access' button. A notification bubble indicates 'Console sign-in link copied'.

The screenshot shows the AWS Sign-In page for an IAM user. The page includes a header with the AWS logo and a 'Try the new sign in UI' banner. The main content area is titled 'Sign in as IAM user' and contains a form with fields for 'Account ID (12 digits) or account alias' (623339241378), 'IAM user name' (user-3), and 'Password'. There is a 'Remember this account' checkbox and a 'Sign in' button. A sidebar on the right features an 'Amazon Lightsail' advertisement with the text 'Lightsail is the easiest way to get started on AWS' and a 'Learn more' button. The bottom of the page shows the Windows taskbar with various application icons.

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The screenshot shows the AWS Management Console for the us-east-1 region. The 'Instances' page displays a table with two instances:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
LabHost	i-0e5b04b2a8d9f560c	Running	t2.micro	2/2 checks passed	User: arm:aws/	us-east-1a	ec2-52-91-
Bastion Host	i-099a75241c8af9ece	Running	t2.micro	2/2 checks passed	User: arm:aws/	us-east-1a	ec2-54-19-

The 'LabHost' instance details are expanded, showing it is in a 'Running' state. Key details include: Hostname type (IP name: ip-10-1-11-154.ec2.internal), Private IP DNS name (ip-10-1-11-154.ec2.internal), Instance type (t2.micro), VPC ID (vpc-0a4db46dd5248d273), and Auto-assigned IP address (52.91.218.124 [Public IP]).

The screenshot shows the AWS Management Console after the 'LabHost' instance has been stopped. A green banner at the top indicates 'Successfully initiated stopping of i-0e5b04b2a8d9f560c'. The 'Instances' table now shows the 'LabHost' instance in a 'Stopping' state:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
LabHost	i-0e5b04b2a8d9f560c	Stopping	t2.micro	2/2 checks passed	User: arm:aws/	us-east-1a	ec2-52-91-
Bastion Host	i-099a75241c8af9ece	Running	t2.micro	2/2 checks passed	User: arm:aws/	us-east-1a	ec2-54-19-

The 'LabHost' instance details are expanded, showing it is in a 'Stopping' state. Key details include: Hostname type (IP name: ip-10-1-11-154.ec2.internal), Private IP DNS name (ip-10-1-11-154.ec2.internal), Instance type (t2.micro), VPC ID (vpc-0a4db46dd5248d273), and Auto-assigned IP address (52.91.218.124 [Public IP]).

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Lab 1: Introduction to AWS IAM

AWS Identity and Access Management (IAM) is a web service that enables Amazon Web Services (AWS) customers to manage users and user permissions in AWS. With IAM, you can centrally manage **users**, **security credentials** such as access keys, and **permissions** that control which AWS resources users can access.

Lab overview and objectives

This lab will demonstrate:

```
graph TD; Account[AWS Account] --> Users[Users]; Account --> Groups[Groups]; Users --> user-1[user-1]; Users --> user-2[user-2]; Users --> user-3[user-3]; Groups --> EC2-Admin[EC2-Admin]; Groups --> EC2-Support[EC2-Support]; Groups --> S3-Support[S3-Support]; EC2-Admin --> IAM-Policy-EC2-IAM[IAM Policy EC2 IAM Start and Stop Access]; EC2-Support --> IAM-Policy-EC2-IAM-ReadOnly[IAM Policy EC2 IAM ReadOnly Access]; S3-Support --> IAM-Policy-S3-ReadOnly[IAM Policy S3 ReadOnly Access];
```

Lab 1: Introduction to AWS IAM

AWS Identity and Access Management (IAM) is a web service that enables Amazon Web Services (AWS) customers to manage users and user permissions in AWS. With IAM, you can centrally manage **users**, **security credentials** such as access keys, and **permissions** that control which AWS resources users can access.

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```
graph TD; Account[AWS Account] --> Users[Users]; Account --> Groups[Groups]; Users --> user-1[user-1]; Users --> user-2[user-2]; Users --> user-3[user-3]; Groups --> EC2-Admin[EC2-Admin]; Groups --> EC2-Support[EC2-Support]; Groups --> S3-Support[S3-Support]; EC2-Admin --> IAM-Policy-EC2-IAM[IAM Policy EC2 IAM Start and Stop Access]; EC2-Support --> IAM-Policy-EC2-IAM-ReadOnly[IAM Policy EC2 IAM ReadOnly Access]; S3-Support --> IAM-Policy-S3-ReadOnly[IAM Policy S3 ReadOnly Access];
```

Task	Score
Total score	40/40
TASK 2a - Added user-1 to S3-Support group	5/5
TASK 2b - Added user-2 to EC2-Support group	5/5
TASK 2c - Added user-3 to EC2-Admin group	5/5
TASK 3a - user-1 logged in	5/5
TASK 3b - user-2 logged in	5/5
TASK 3c - user-2 ec2 stop instance attempt	5/5
TASK 3d - user-3 logged in	5/5
TASK 3e - user-3 EC2 stop instance attempt	5/5

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The screenshot shows the AWS Academy course page for 'Introduction to AWS IAM'. The course is titled 'ACFv2EN-LTI13-85822' and is part of the 'Grades' section for 'Dwivedi, Divya'. The page displays a table of grades for various activities and labs. The total score is 100%.

Name	Due	Submitted	Status	Score
Activity - AWS Elastic Beanstalk Labs				- / 100
Activity - AWS Lambda Labs				- / 100
Course Assessment Knowledge Checks				-
Lab - 1 Introduction to AWS IAM Labs	Sep 20 at 8:22pm			100 / 100
Lab - 2 Build your VPC and Launch a Web Server Labs				- / 100

On the right side, there is a summary section showing 'Total: 100%' and a 'Show All Details' button. Below this, it states 'Course assignments are not weighted.' and a checkbox for 'Calculate based only on graded assignments' which is checked. A note explains that grades are based on What-If scores and that test scores for assignments already graded are included.

The screenshot shows a detailed breakdown of grades for the 'Introduction to AWS IAM' course. The page displays a table of grades for various modules, knowledge checks, assignments, and labs. The total score is 100%.

Module	Score
Module 5 Knowledge Check Knowledge Checks	- / 100
Module 6 Knowledge Check Knowledge Checks	- / 100
Module 7 Knowledge Check Knowledge Checks	- / 100
Module 8 Knowledge Check Knowledge Checks	- / 100
Module 9 Knowledge Check Knowledge Checks	- / 100
Module 10 Knowledge Check Knowledge Checks	- / 100
Assignments	N/A 0.00 / 0.00
Knowledge Checks	N/A 0.00 / 0.00
Labs	100% 100.00 / 100.00
Total	100% 100.00 / 100.00